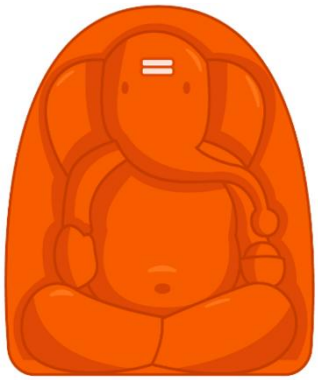






Computer Algorithm

Tutorial No. 1

1. What is an algorithm and give its characteristics.
2. Explain different asymptotic Notations.
3. Explain performance analysis and performance measurement.
4. Explain Tower Of Hanoi problem in detail and analyze its complexity.
5. Compare the complexities of n, logn,nlogn,n^2,n^3,2^n.
6. The factorial function n! Has value 1 where n<=1 and value n*(n 1) when n>1,write both recursive and iterative algorithm t compute n!.
7. The Fibonacci numbers are defined as f0=0,f1=1 and fi=fi-1+fi-2 for i>1. Write both recursive and iterative algorithm to compute fi.

Tutorial No.2

1. Explain divide and conquer algorithm with recurrence relation.
2. Show that number of comparisons for finding maximum and minimum is 3n/2-2 where n is the no. of elements.
3. Explain recursive binary search algorithm with example.
show that complexity of binary search is O(logn) for successful search
4. Explain merge sort algorithm with example and calculate its complexity.

. Explain quick sort algorithm with example and and show that its complexity is O(nlogn).

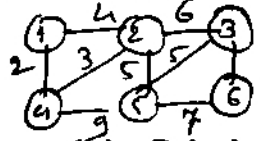
. Explain selection sort algorithm and show that its complexity is)(n^2). or write the algorithm to select nth smallest element in the list & compute it's complexity

Tutorial N0.3

. Write a short note on Greedy method.

. Find an optimal solution to knapsack instance n=6,c=20,w=(w1,w2,w3,w4,w5)=(2,3,5,7,1,5) and profit is(p1,p2,p3,p4,p5)=(12,5,15,7,6,18).

. Prove that if P1/W1 >=P2/W2 >= ...>= Pn/Wn then Greedy method generates an optimal solution to the given instance of knapsack problem.



. Define Spanning tree with suitable example and explain Prim's algorithm to find Minimum Cost Spanning Tree.

. Explain Kruskal's algorithm and compute Minimum Cost Spanning tree with example.

Solve job sequencing with deadline problem using greedy approach for following problem instances.
n=5, (P1, P2, P3, P4, P5) & (d1, d2, d3, d4, d5)
45 15 55 65 75 1, 3, 2, 1, 2.

Tutorial N0.4

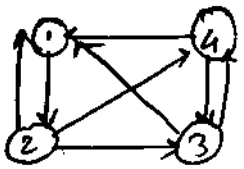
1. Write a short note on Optimal Storage on Tapes.

2. Obtain a Huffman code for message (m1,m2,....,mn) with relative frequencies (4,5,7,8,10,12,20). Draw decode tree for this set of codes.

3. Find an optimal binary merge pattern for 10 files whose lengths are 28,32,12,5,84,53,91,35,3.

4. Write a note on single source shortest path algorithm.

Tutorial No.5



$$\begin{bmatrix} 0 & 5 & \infty & \infty \\ 50 & 0 & 15 & 5 \\ 30 & \infty & 0 & 15 \\ 15 & \infty & 5 & 0 \end{bmatrix}$$

1. What is a multistage graph?

2. Explain all pair shortest path using dynamic programming.

3. Obtain Optimal binary search tree for identifiers set A1,A2,A3=(do,if,while) with probability p1=0.5,p2=0.1,p3=0.05 and q0=0.15,q1=0.1,q2=0.05,q3=0.05.

4. Explain 0/1 knapsack problem with example in dynamic programming.

5. Explain with example Travelling Sales Person problem.

6. Solve the following instances of reliability design problem with 3 stages and cost of system is 105.Cost of devices of stage 1 is 30,stages 2 is 15 and for stage 3 is 20.Reliabilities of stages are 0.9,0.8 and 0.5 respectively.

Tutorial N0.6

1. Explain binary tree traversal techniques with example.

2.Explain searching techniques for graph.

3.How can be a non connected graph is converted into connected graph. Define an articulation point how non connected graph can be converted to bi connected component.

4.Write a short note on

1. Game Tree
2. And /OR Graph

Tutorial N0.7

1. Draw and Explain a permutation tree for 4 queen problem.
   Explain 4queen Problem (write algorithm to test no 2 queens are placed in the same diagonal, row & column.
2. Explain Sum Of Subset problem using backtracking.
3. Explain Hamilton cycle with suitable example.
4. Explain Graph Coloring problem using backtracking.
5. Expalin the following terms with suitable example.
   1. Live Node
   2. E Node
   3. Dead Node
   4. Bounding Function.

Tutorial N0.8

1. With neat diagram explain relationship between P,NP,NP-hard ,NP-Complete.
2. List and Explain NP-hard problems. → graph
   clique Decision Problem
   Node cover decision Problem
   Directed Hamiltonian cycle.
   chromatic number Decision Problem.
3. Define the following terms

1. Deterministic Algorithm

2. Non –Deterministic Algorithm

Tutorial N0.9

1. Describe and give its example of prefix computational model with PRAM.

2. List and Explain variants of PRAM.

3. Explain 3D Butterfly network for hypercube.

4. Explain Odd Even merges on mesh with neat diagram.

5. Broadcasting with mesh & hypercube

6. Prefix sum computation in hypercube.

7. Prefix sum computation with mesh & hyperbe

8. what is a determinstic list ranking problem in PRAM, Explain with example.

9. Explain PRAM computational model

10. Explain 'packet routing in square & mesh & linear array.

11. Data concentration with mesh & hypercube.

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3. Cloud Computing platform & cloud Native technology

cloud computing utilizes the frame work that is build for hosting extensive distributed network applications on the internet.
cloud architecture enables the integration of software operating on virtualized hardware across several location to deliver a user sending service on demand.

composability :

cloud based applications typically causes the characteristics of being from a set of components which is known as composability.
A composible system utilizes components to provide services that can be customized for a specific purpose using standardize elements a composible component must possess a following characteristics.

(1) Modular :
A modular unit is a self content and independent component that is operative reuserable and replacable.

(2) stateless :
A transaction is conducted independent of other transaction or request.

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Infrastructure :

The majority of major IaaS providers depend on virtual machine technologies to supply services capable of running application. The VMM component often known as Hypervisor the VM monitor these software is responsible for enabling several OS to operate independently in their own allocated memory and handle I/o operations for the VM.

platforms :

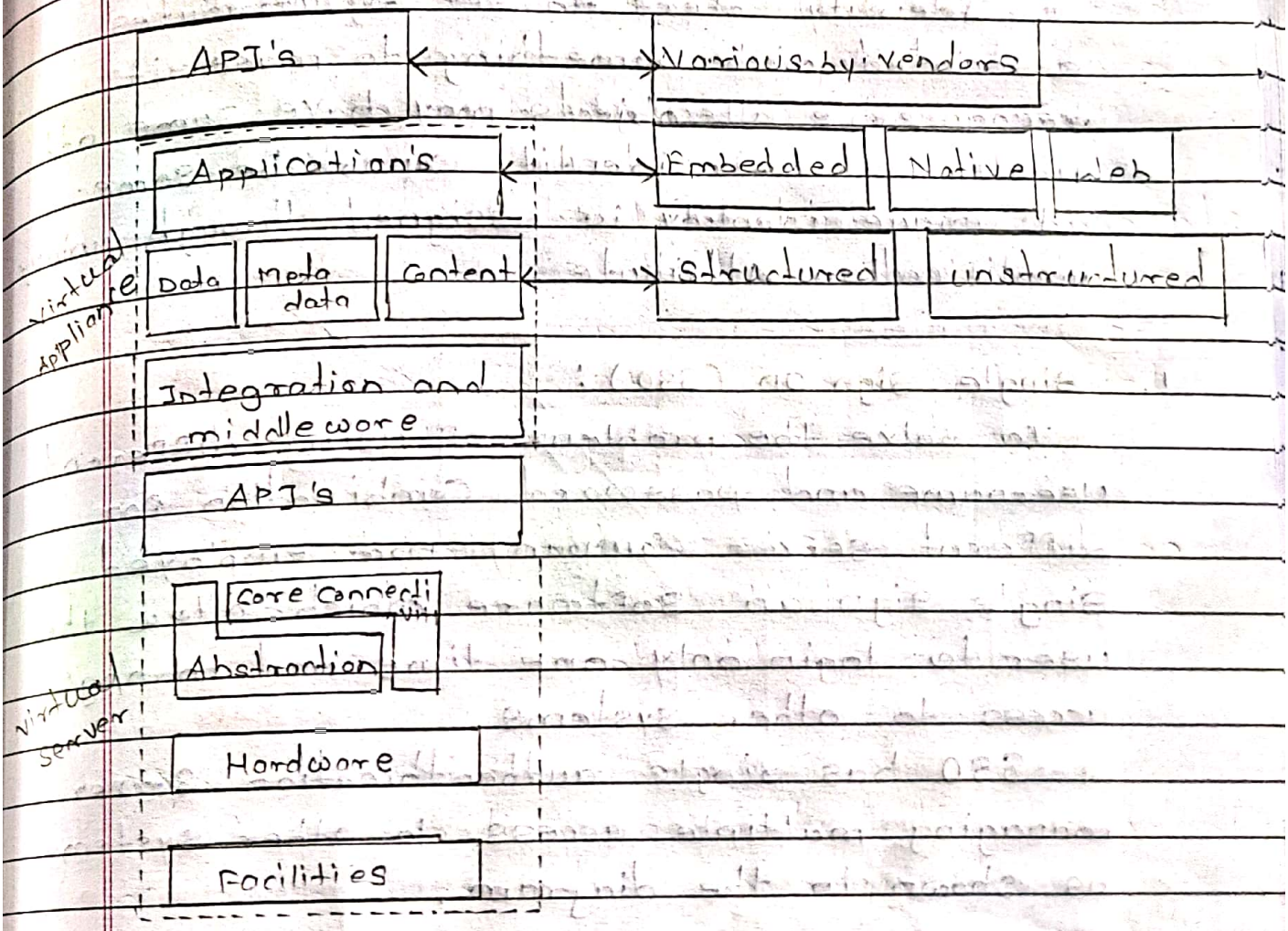
A cloud platform is a software layer that enables the creation of more advance services, various PaaS, Vendors often services design to give developers with diverse capability. Platforms encomposed all most the whole cloud software stack with the exception of the presentation layer responsible for the user interface. platforms frequently include a wide range of tools and utility that assets in the process of desigining and deploying application.

Various vendors often a range of developer tools that kater to deal communication, testing, program performance measurement, attribute analysis, version control, database and web service integration and storage management typically platforms initates the process by creating or developer Community to provide assitants and support for the task

performed within the system.

Virtual appliances :

Virtual appliances are applications such as web servers or database servers that can be executed on VM image. A Virtual appliances a software programs that are loaded on virtual servers they are desiged to execute on specific machine instances or image type. A virtual appliance is an instance of a platform virtual appliances are progression on the middle of cloud applications.



Identity as Service :

Employeems in a Company require to login to system to perform various task these systems may be based on local servers or cloud based servers. following are the problems that an employee might face :

1. Remembering username and password Combination for accessing multiple servers.
2. If and employee leaves a Company it is required to all the accounts disable.

To Slove these problems the new technique Identity as a service is used.

Identity Refers to set of attributes associated with something to make it recognizable all objects may have same attributes but their identity cannot be same. A unique identity is assigned through unique identification Attribute.

1. Single Sign On (SSO) :

To Solve the problem of using different username and password Combination for different server Company now employed Single sign on software which allow the user to login only one time and manage access to other systems.

SSO has single authenthication server managing multiple access to other system as shown in the diagram.

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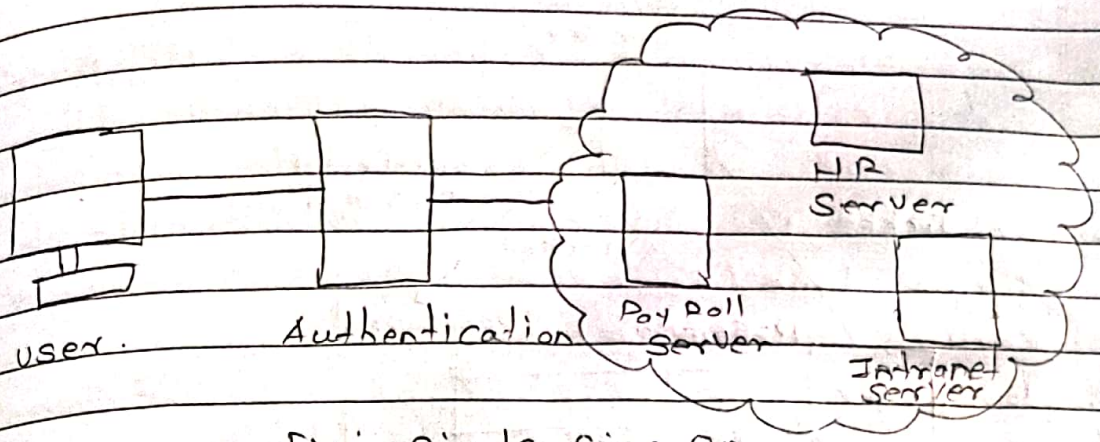


Fig: Single-Sign On

Working of Single Sign On:

Their are several implementation following steps explain the working of SSO software.

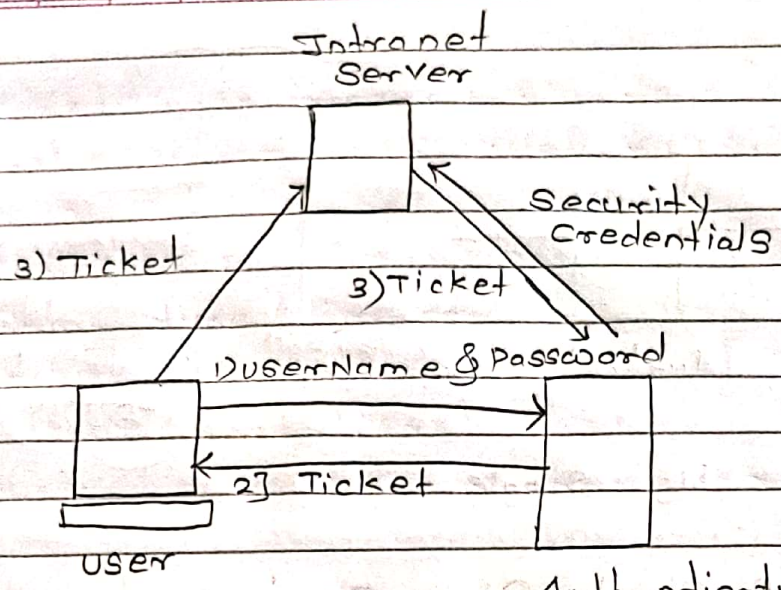
Step 1:
User logs into the Authentication Server using a username and password.

Step 2:
The authentication Server returns the users tickets.

Step 3:
User Sends the ticket to intranet server both intranet Server sends that ticket to the authentication Server

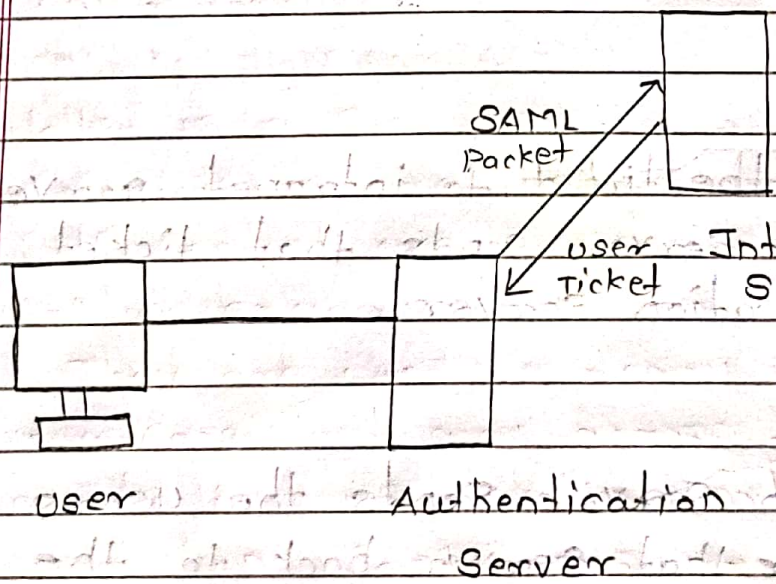
Step 4:
Authentication Server Sends the users security credintals for that Server back to the intra-net Server.

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2. # Federated Identity Management (FIDM):

FIDM discribes the technology and protocols that enables user to package security credentials accross security domain. It uses SAML (Security Markup language) to package a users Security Credentials as shown in the diagram.



8. Open ID :

IF offers users to login in into multiple websites with single accounts. Google, MySpace, blogbrase.com, Yahoo are some of the companies that supports open ID.

Benefits :

1. increase side conversation Rate
2. Excess to greater user profile Content
3. fewer problem with lost passwords
4. Is of content integration into social networking sites

Compalince as a Service :

CaaS is a cloud based delivery model where third party providers or manage service providers automate, manage and monitor and organization association to industry and goverment guidelines regulation.

It performs manuals periodic audits into Continous automated workload.

key Features :

1. Continous monitoring : Real-time tracking of cloud resource configuration and security Controls.
2. Automated Evidence Gallery : Instead gathering of reports and documents required for audit.
3. Policy Maping : Translating Complex laws into technical clouds cloud Configuration.

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4. Risk Management : Early detection of miss-
5. Configuration that voilet rules.

Benefits :

1. Same type / time
2. Reduce human error
3. lowers operational cost
4. Speed up audits preparation

Drawbacks :

1. Creates reliance on a third party vendor
2. offers limited customization for unique work flow
3. The ultimate legal responsibility for a beach still reach on a business.

Containers Vs Virtual Machines :

Containers and VM are two technologies that enable application to run in isolated environments.

What is a Container :

Containers are light weight, portable and self-Content executable image that Contain Software applications and their dependancies

They are used to deploy and run applicatio in a Consistant way across different environments such as development, staging and production. Containers are typically deploye from an image by using an orchestration platform like Ubernets.

What is Virtual Machine:

VM, or guest represent instance of an OS colocated on a physical machine through the use of a hypervisor.

Each VM has its own OS, memory and other resources which are isolated from the other VM's on the same physical Computer this allows multiple OS to run on the same physical Components without interferring with each other.

Difference between Container and VM.

Virtual Machine	Container
VM excess the hlw of physical machine through hypervisor	

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	Feature	Container	VM
1.	Virtualiz-ation level	OS level	hardware level
2.	OS	shares host OS kernal	Guest OS
3.	Size	Mega bytes	Giga bytes
4.	boot type	Seconds	Minutes
5.	Resource usage	lower	higher
6.	isolation	process level	hardware level
7.	Protability	high	low
8.	Management	typically managed by Container orche-stration.	Managed by hypervisor.

To understand the difference between Container and VM it is helpful to visualize their architecture.

1. Container:

1. Share the host kernal : Containers levarage the underlined OS kernal of the host machine

2. light-weight :
   Smaller in size and require few resources

3. Faster startup:
   Starts and stop quickly due to shared OS.

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2. Virtual Machine:

1. Runs a full guest OS : each VM has its own dedicated OS.
2. Resource extensive : Require more resources CPU, memory, storage
3. Slower startup : take longer to boot due to the need to load the guest.

- Similarities between Container and VM's :

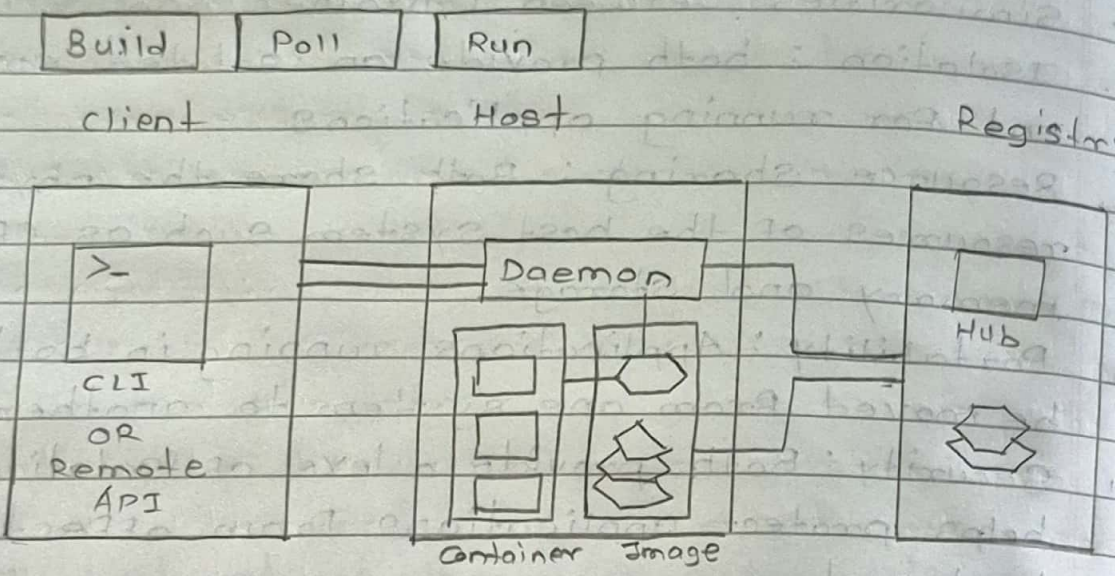
1. Isolation : both provide an isolated environment for running applications.
2. Resource sharing : Both share the physical resources of the host system such as CPU, memory and storage.
3. Portability : Applications running in both can be moved from one system to another.
4. Security : Both provide a level of isolation that helps protect applications from affecting the host system.
5. Deployment : Both are used to simplify and speed up application deployment.
6. Scalability : Both can be created managed and scaled according to application requirements.
7. Environment Consistency : Both help ensure that applications run in a consistent environment across different systems.

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Docker :

A docker is a open source platform that enables developer to package, application, and their dependancies into portable containers that can run consistently across various environment.

Docker Architecture :



Docker uses a client server architecture following is a breakdown of the core components associated with docker.

1. Docker host :

A docker host is a physical or VM running linux or another docker agent compatible os

2. Docker engine :

docker engine is a client server application consisting of the docker daemon, a docker API that entracts with b. and CLI that

docks to daemon.

3. Docker daemon:
Docker daemon is a service that creates and manages docker images by using the commands from the line s the docker for docker daemon server as the docker implementation.

4. Docker client:
The docker client provides the CLI that excess the API to communicate with the docker daemon over unique socket or a network interface. the client can be connected towards the daemon and developer that run the daemon on the same computer system

5. Docker Container
Docker Containers are the life running instances of docker image by docker images are read only files.

6. Docker image
docker image contain executable application source code and all the tools, libraries and dependancies the application code needs to run as a container.

7. Docker build
Docker build is a command that has tools and feature for building docker image.

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8. Docker Files :
Docker Files, each docker Container starts with a simple text file Containing instructions for how to build docker Container image

9. Docker hub :
docker hub is the public repositary of docker images. docker hub includes images produced by docker certified images belonging to the docker and thoustand of other images.
docker resister is open source Scalable and distribution system for docker image.

Benefits :
Docker has revolutaries the way we develop, shift and run applications by enabling Containraition the key benefits of docker are.,
1. portability - Consist environment docker ensures that application runs the same way of first different enviironment.
2. Cross platform Compatible - docker Container can run on any system that supports docker.
3. Effiency - light weight : docker Container Share the host system os kernel making them more light weight.
- Fast - startup time - Containers can be starts and stop quickly
4. Scalability and performance
- Efficient resource utilization - unlike VMs Container do not require a full OS to be replicated.

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- Easy Scaling - Scaling can be done easily by running multiple instance on computer.

5. isolation
   - Application isolation - Each application runs in own isolated environment and do not interfare with each other.
   - Process isolation - processes running inside a container are isolated from those running on the host or other containers.
6. Continous integretion / deployment
   - integretion with CI/CD tools -
   - Rapid and reliable deployment

3. ConfigMap:
- for basic changes like URL changes could be require to change in application properties
- the repeat the image.
- Push the changes to the it environment.
- Duber nets provides us a ConfigMap which
- can do this easily

4. Secret:
- It is used to store secret data.
- It is in base 64 incoding

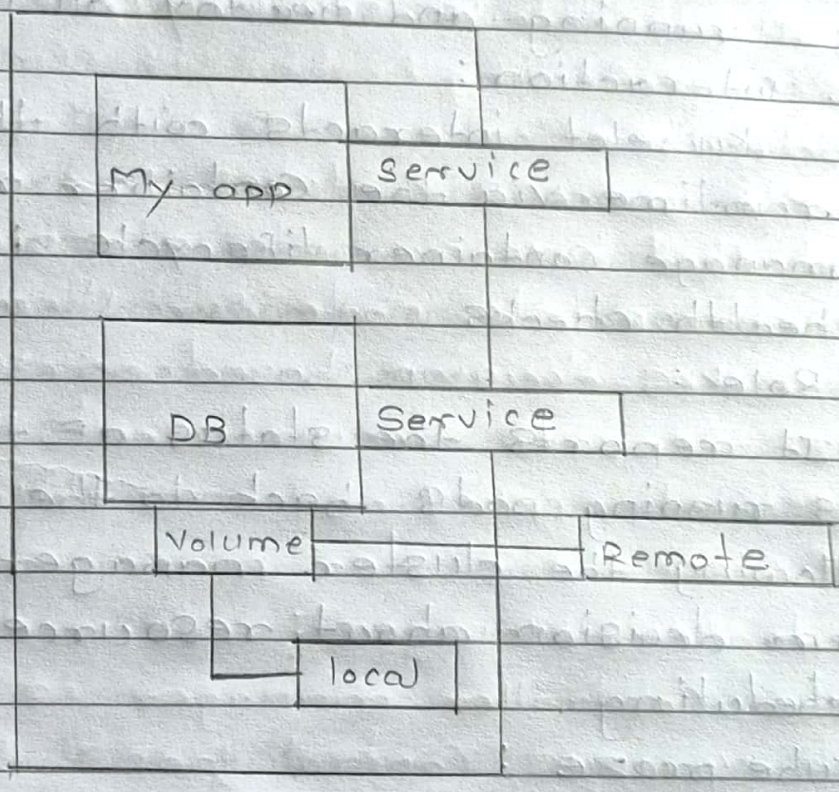
5. Volumes:
- If your database crashes data is lost.
- Duber nets does not manage data persis

We can use volume as a local machine or external hard drive. being as users are responsible for peristency and backup.

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6. Deployment:
- If our database or application crashes their could be down time.
- In this case we need to specify the No. of backup or Container we want
- The Service will point to the same IP address.
- Helps in load balancing.

7. Statefull Set:
- We can deploy database directly because multiple Containers could be written to the same database For this we manage a StatefullSet.



Kubernetes Architecture:

1. Worker Node and processes:
   We need three processes for the worker machine in Kubernetes cluster to work

2. kubelets:
   The kubelet is the primary node agent that ensures Containers are running in a pod as specified by the Kubernets Control claim
   1. Functions:
      - It receives instructions from the API Server and ensures that the Container describe in those specifications are running and healthy on its node
   2. Interaction:
      Kubelet interacts with the Container runtime via the Container runtime interface manage Container lifecycle image and health checks
   3. Role:
      It reports the status of the node and its running pods back to the Control plane allowing the cluster manager to make informed decision about resource allocation and scheduling.

4. Kube proze:
   Kube proze is a netcoork proze that runs on each node to implement kubernets Service Concept
   1. Function:

It maintains network rule to allow network Communication to and from pods within the cluster.

2. Role:

It manages cohole forwarding and simple load balancing for services, ensuring that traffic through a stable service IP is routed to the correct dynamically changing pod IP.

4. Container Runtime:

The Container Runtime is the underlined Software that is responsible for executing Containers and managing container image on a node.

1. Function:

It puts Container image from a resistant unpack them and runs the application within an isolated environment managing resources like CPU and memory using mechanism like C-groups and namespaces

2. Role:

It provides the necessary environment for the application workload to run in Scheduling and efficiently.

5. Master Node and processes:

The kubernets master node runs several core processes responsible for managing and maintaing the desire state of a entire cluster. each Components acts as the brain

of the cluster while worker nodes run the actual application the main master processes are, API server these is the frontend of the kubernets control plane and the single entry point for all communication it exposes the kubernets API, process reset request, validates them and stores the Configuration data in ETCD.

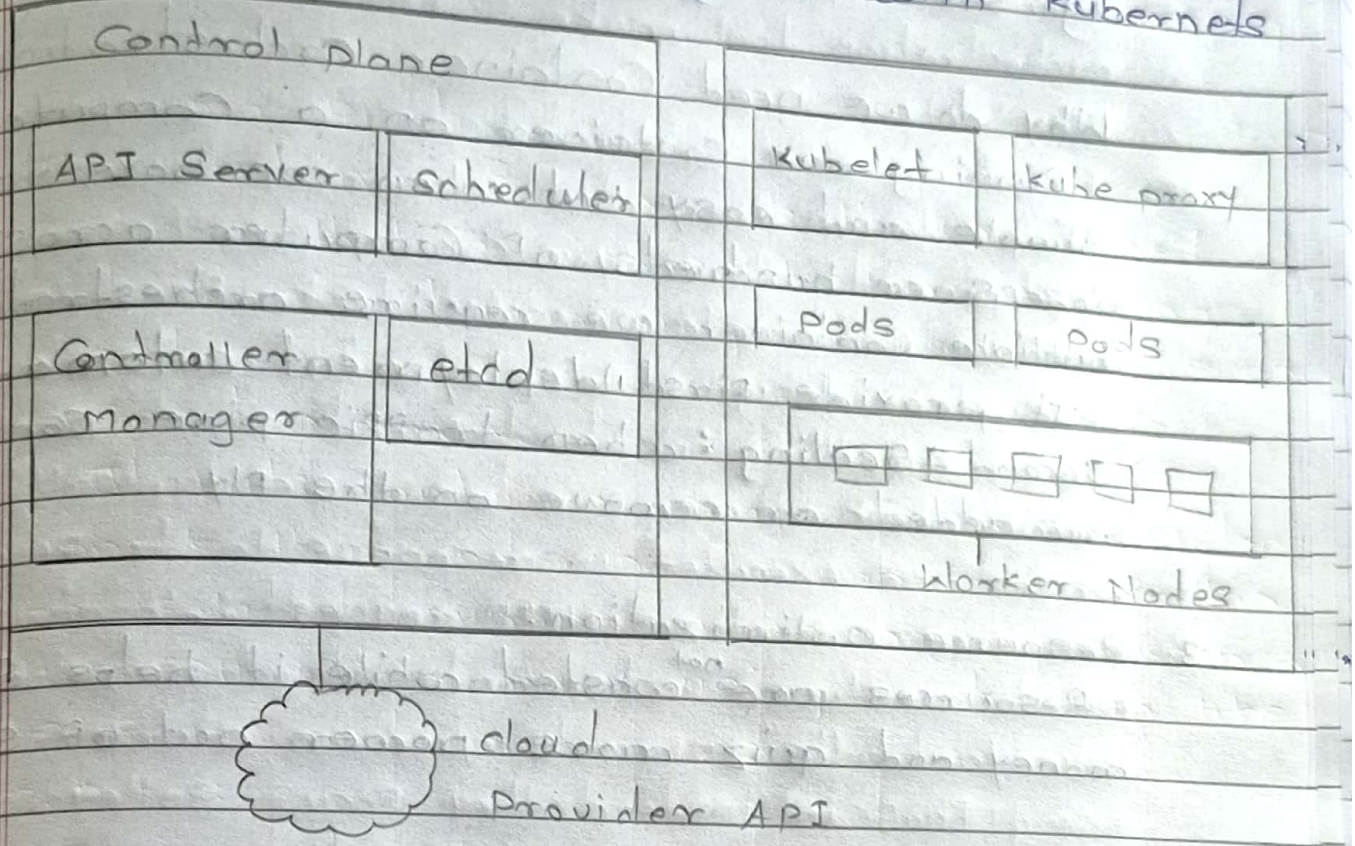
6. ETCD :
A consist and highly available key value store that acts as the clusters single source of cluth. it stores all data Configuration and the current state of the cluster such as pod details services and node status

7. Schedular :
The process watches for newly created pod that have no node assigned and selects an appropriate node for them to run on it makes decision based on various factors like resource requirements, hardware Constaints, policy and availability

8. Controller Manager :
kube Controller manager is a deman that runs various Contral loops that Continously monitor the actual state of the cluster via the API Server and worth to move it towards the desire state.

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9. cloud Controller manager:
This Component is used in cloud environments to manage interaction with the underline cloud providers API for task such as managing load balancers and storage volumes. it only runs Controllers specific to the cloud provider allowing cloud infrastructure to integrate with kubernets



Container Orchestration:
Container orchestration means lating an automated system handle all the heavy lifting when it Comes to managing Containers. Container comps up everything and application needs into need easy to move package the orchestration tool make suru all these Containers get where they need to go,

start when they shift, Scale up and down as needed and stay connected No matter what you can picture containers are the shipping grops holding application and its dependencies while orchestration system is like a port manager keeping track of each Container moving them as needed and making Sure nothing gets lost or stuck.

Why do we need Container Orchestration:
Running one Container on a Computer is simple and easy but when we have to dozens or hundreds of Computer across multiple Container we require orchestration.
It poroide first 1) **deployment automation**: Second 2) **Scaling**: when traffic pipe Containers are added or remove on the fly

3. Resource optimization:
Resources are not wasted while it helps Save money and quiz more power out of Server

4. Self cleaning:
If Something fail the System replaces it did not mean

#5 Networking:
With it become Smooth and Safe Secured between Container.

6 Visibility:

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orchestration gives all the log and monitoring so that you know how your app are to be.

How Container orchestration platform works.

1. Deployment and Scheduling:
The orchestration tool chooses which server runs each container these server effort worker node whenever new application is pushed the system desires where to put each piece may be start with everything on one node as the application grow add new worker node to spread the load the system balancing every thing for the app to be speedey and reliable

2. Scaling application:
Whenever the traffic pipes up if the front end needs handle more users add more Containers if the backend or database stuck you can scale them

3. Networking and Service discovery:
Orchestration creates a network getting containers to talk to each other only Containers we want are visible from outside internal services have stable address even if Containers move around.
Node balancing happens automatically.

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4. Monitoring self cleaning and inside:
local Containers are replaced automatically if a node dies the Container moves healthy Somewhere while it provides logs and monitoring.

5. Declarative Configuration:
You can write whatever you want while the orchestration tool make sure happens.

Function as a Service:
FaaS is an important approach to Software development where you Create small Specified function to handle Specific task. FaaS is model in system design where developers write and deploy small independent functions to perform Specific task these functions are typically event driven and executed in response to the Reverse or request such as HTTP request, database events or time coast events. developer don't have to handle infrastructur or Servers by using FaaS int instead they Focus Solley on writing Code for indivisual function.

The FaaS platform provides and Scales the underline resources automatically based on the demand it is key Component of Serverless architecture also known as Serveless Computing or FaaS is the cloud Computing model ~~where~~ were cloud providers, dynamically manage the allocation and provising and of server

enabling programmers to concetrate only on creating and implementing functions as code.

Key Concepts of Faas:
This are revolves around principle and components as follow:

1. Function:
Functions are building block of Faas they are small independent unit of design to perform specific task or respond to event.

2. Event driging:
These functions are by events such as HTTP request, database changes, File upload or timeup the treger initate execution of corresponding

3. Serverless:
Faas abstracts the need to manage server or infrastructure develeoper do not need to provising scale and maintain server thus the platform handles the task automatically allowing serverless Computing model.

4. Pay Paper use:
Faas follows a paper use within model coere users are charged based on the resources Consumed by their functions.

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5. Scalability :
FaaS platform automatically scales function it respond to work load demand it handle different level of

Benefits :
1. worst Scability
2. Her Cost efficiency
3. Simplicity
4. Flexibility
5. Event repeat Architecture
6. High availability

- FaaS Architecture and its Component :
The architecture consist of following Componen
1. Function
2. FaaS provider
3. Event resources
4. Function execution environment : Each Funct runs in its own isolated execution environme which includes the neccessary runtime environ and HTTP code the execution environment is provising dynamically when function is invoke and discard it after execution to conserve resources.
5. Invokation Resources
6. Monitoring and login

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Unit : 3

Cloud computing platform and cloud native technologies

- Cloud computing utilizes:

Cloud computing utilizes the framework i.e build for hosting extensive distributed network applications on the internet.

Cloud Architecture enables the integration of software operating on virtualize hardware across several location to deliver a user centric service on demand.

Compability:

Cloud based applications typically the characteristics of being from a set of components which are known as compatibility. A composible system utilizes components to provide service that can be customize for a specific purpose using standardized elements. A composible component must the following characters.

1) Modular

A modular unit a self contain and independant component i.e. composable, reusable and replacable.

2) Stateless

A Transaction is conducted independantly of other transaction or request.

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Infrastructure

The majority of major infrastructure and service providers depend on virtual machine technologies to supply services capable of running applications.

The virtual machine component of a hypervisor is the virtual machine monitor. This software is responsible for enabling several or. To operate independently own allocated memory and handles Input / output for the virtual machine.

Platforms:

A cloud platform is a software that performs the creation of more advance services various platform as a services vendors offer services design to give developers with device capability.

platform encompasses almost the whole cloude services with the exception of the presentation layer responsible for user interface

platform frequently includes a wide range of tools and utilizes that assist in the process of designing and deploying applications various vendors offer to team, communication testing program performance measurement attribute analysis database and webservice integration and storage management.

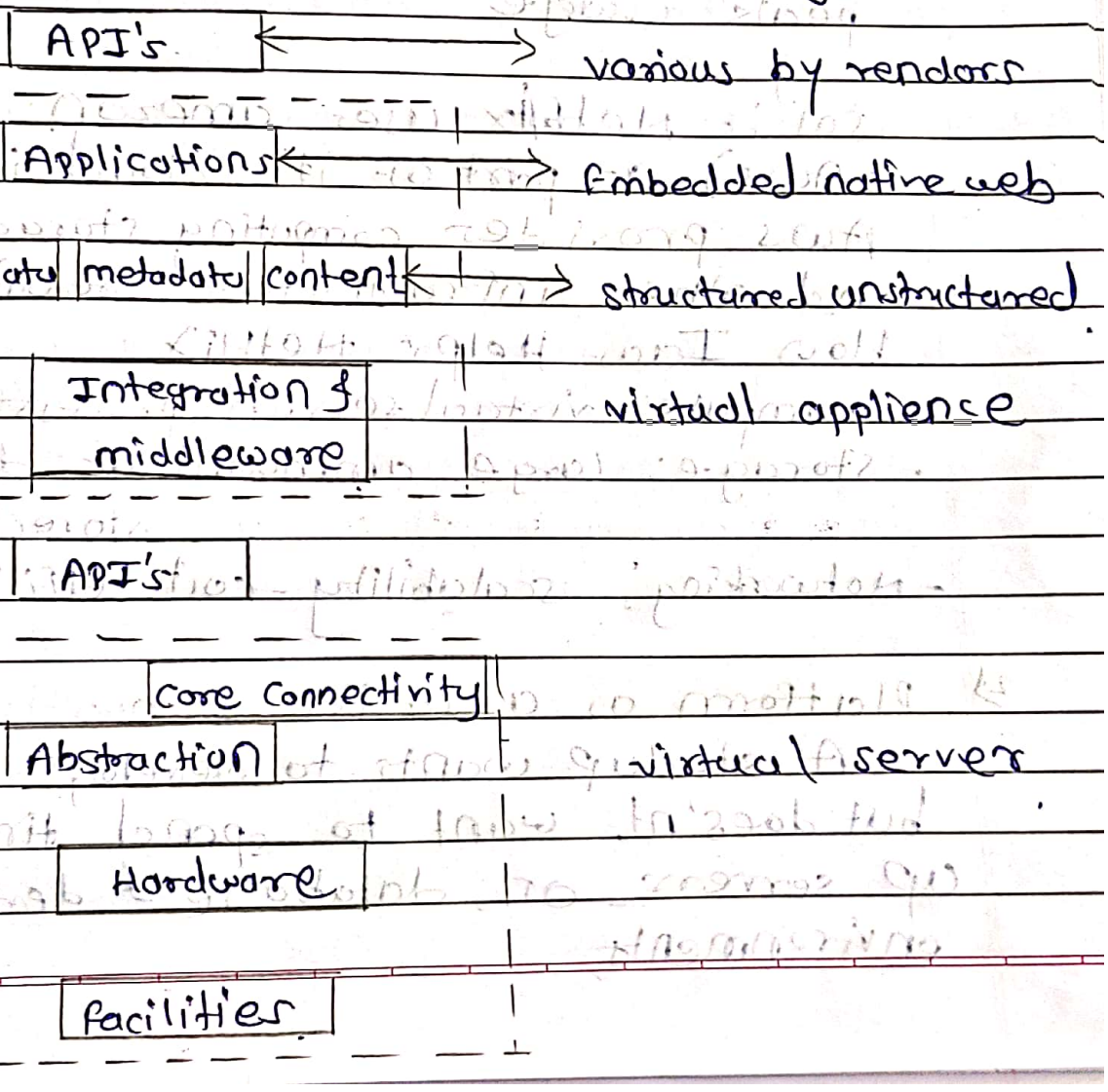
Typically platform is initiate the process by creating a developer community to provide assistance and support for the task platform within the system.

Virtual appliances:

it is are applications such as web servers or database servers that can be excited on a virtual machine image

A virtual appliances are software program that are loaded of virtual servers. They are design to execute on specific machine instances or image type

A virtual appliances is an instance of a platform Virtual appliances are position in the middle of Cloud computing.



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Platform as
- Case study on

1) Infrastructure as a service
Netflix is popular online streaming platform that provides movies and TV shows to millions of users worldwide. Handling such a large number of user requires huge computing storage and networking resources.
problem:
Netflix needed:
- large amount of storage for video
- powerful servers to handle midleince of users.
- Ability to increase resources during peak usage.

sol^n:- Netflix uses amazon web services (AWS) as part of its cloud infrastructure. AWS provides computing storage networking and other infrastructure resources on demand.
How Iaas Helps Netflix
- compute: virtual servers process application
- storage: large ammount of data can be stored
- Networking: scalability - cost efficiency

2) Platform as a service
A startup wants to develop a web appl but doesn't want to spend time setting up servers or, database & development environments,

problem:
-Developers traditionally have to set up servers, Install os, configure development environments manage scaling, maintain infrastructure.

sol^n. The company uses google App engine a PaaS offering from google cloud.

3) Software as a service

A college or company needs application email, document creation spreadsheets pentation, and collaboration.

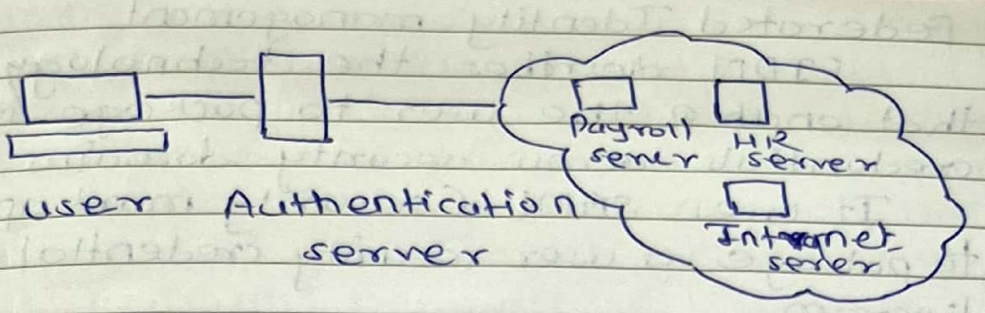
Problem:
Traditionally the organization would be need to purchase software license, Install appl^n on computer, maintain source, perform software updates, manage storage and backups.

sol^n.
The organization uses google workplace which provides application such as gmail google Docs, sheets and slides through the internet.

- Identity as service
  Employers in company require to log in to system to perform various task this systems may be based on local servers or cloud based servers.
  Following other problems that an employee might it face
  i) Remembering username & password for accessing multiple servers.
  ii) if an employee leaves the company it is required that all account to d be disable
  To solve this problem, a new technique called identity as service is been used.

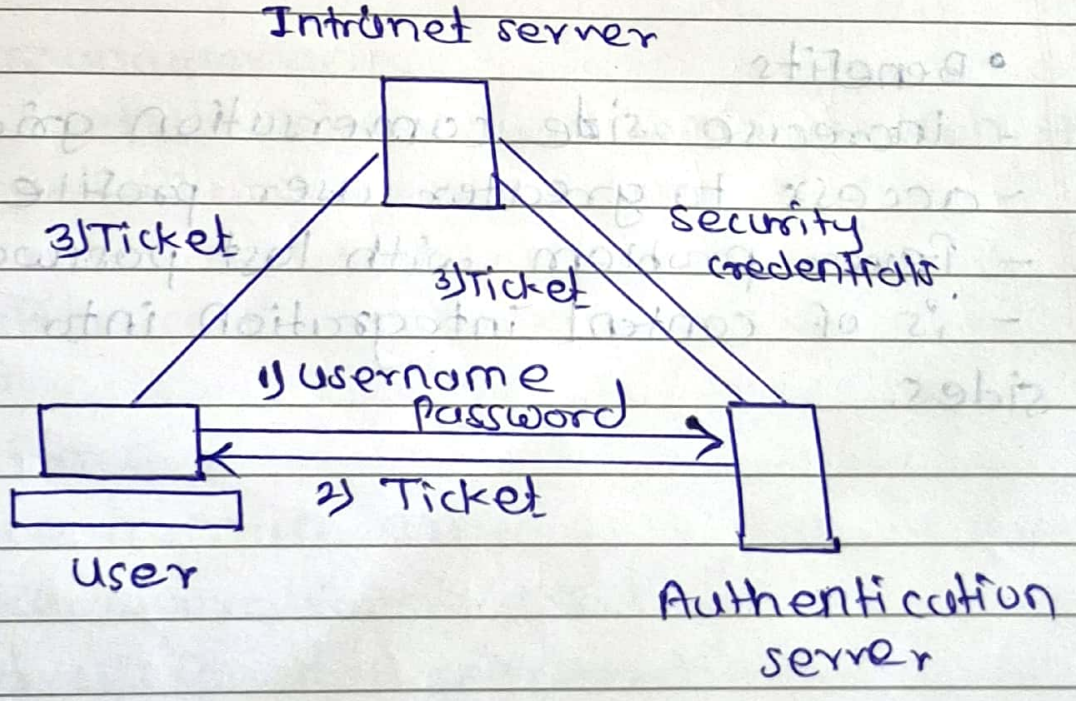
- Identity refers to set of attributes associated with something to make it recognizable. all objects may have same attributes but there identier cannot be same.
  A unique identity is assign through unque identification atribute.

i) single sign on (SSO)
  To solve the problem of using different username & password combination for different servers companies know employee single sign on software which allows the user to log in only one time and manage the access to the other systems.
  - SSO as ~~server~~ single authentication server managing multiple access to other system as shows in diagram.



- SSO working
- Their are separate implementation of SSO following steps explain the working of SSO.

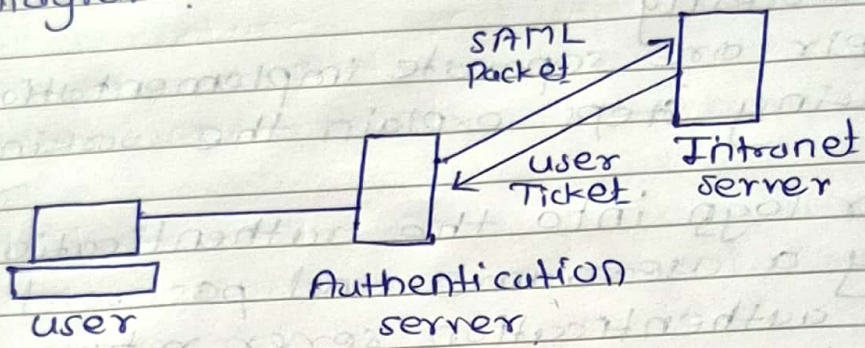
i) user logs into the authentication server using a username and password
ii) The authentication server returns the user ticket.
iii) users sends the ticket to internate server
iv) intranet server to sends the ticket to the authentication server
v) authentication server sends the user security credentials for that servers back to the intranet servers.



2) Federated Identity management

FIDM describes the technology and protocol that enable the user to package security credentials across security domains.

It uses SAML (security mark up language) to package a user security credential as shown in diagram.



3) Open ID

- it offers users to log in into multiple websites with single account.
- Google, Myspace, yahoo, wordpress, dot.com are some of the company that support open ID.

- Benefits
- increase side conversation grid
- access to greater user profile content
- Fewer problem with lost password.
- is of content integration into social networking sides.

- Compliance as service
  - Compliance as service is cloud based delivery model when third party providers or managed service providers automate, manage and monitor an organizations association to industry and government guideliner regulations
  - it performs manuals wi periodic audits into continues automated workflow.

- key features of compliance as service

1) continues monitoring
real time tracking of cloud resource configuration on security controls.

2) Automated evidence gathering
instante of gathering of reports and document required for audit.

3) Policy mapping
Translating complex law into technigual cloud configuration.

4) Risk management
Early detection of miss configuration that violet rules.

- Benefits and droback

- safe time
- Reduce human error.
- lowers operational cost
- speed up audits preparation.

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- Drawbacks
  - creates relience on third party vendor
  - offers limited customization for unique workflow
  - The ultimate legal responsibility for bridge still rest on the business.

- Containers vs virtual machine
  containers and virtual machine are two technologies that enable applications to run in isolated environment

What is container?
- Containers are lightweight portable and self-content executable image that content software applications and they are different dependancy
- They are used to deploy and run applications in consistant way across different environment such as deploy development, staging and production.
- Containers are tipycally deployed from image by using and orchastration platform like google
-

What are virtual machine?
- VM or guest represent instance of an operating system po-located on physical machine through the use of hypervisor.
- each VM has its own operating system memory and other resources, which are isolated from the other VM's on the same physical computer
- This allows multiple operating system to run on the same physical component without interfering with each other

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- Difference bet$. container and VM

i) - VM access the hardware of physical machine through a hypervisor. the hypervisor creates an abstraction layer allowing the VM to access CPU, memory, and storage.
- container on the other hand represent a package that includes an executable with dependancies into needs to run this means that each container shares the physical machin hardware and operating system (kernel) with other containers.

ii) - as a result VM's are typically more resource intensive then containers.

Feature	container	virtual machine
1) virtualization level	operating system level	Hardware level
2) operating system	shares host os kernal	guest os
3) size	megabytes	gigabytes
4) Boot time	Boot time for container second	minutes
5) Resourse usage	lower	Higher
6) Isolation	Process level	Hardware level
7) Portability	Highly portable	less portable
8) management	Typically manage by container orchastration tool	manage by hypervisor

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To understand the difference betⁿ containe and vm it is helpful to visulize there are architecture

Containers
i) shares the host kernal
containers leverage the underline operating system kernal of the host machine

ii) lightweight
Smaller in size and requires less resources

iii) faster startup
start and stop quickly you to shared os kernal

Virtual machine
i) Runs a full guest os
each vm has its own dedicated operating system.

ii) Resource intensive
require more resources that is cpu, memory and storage

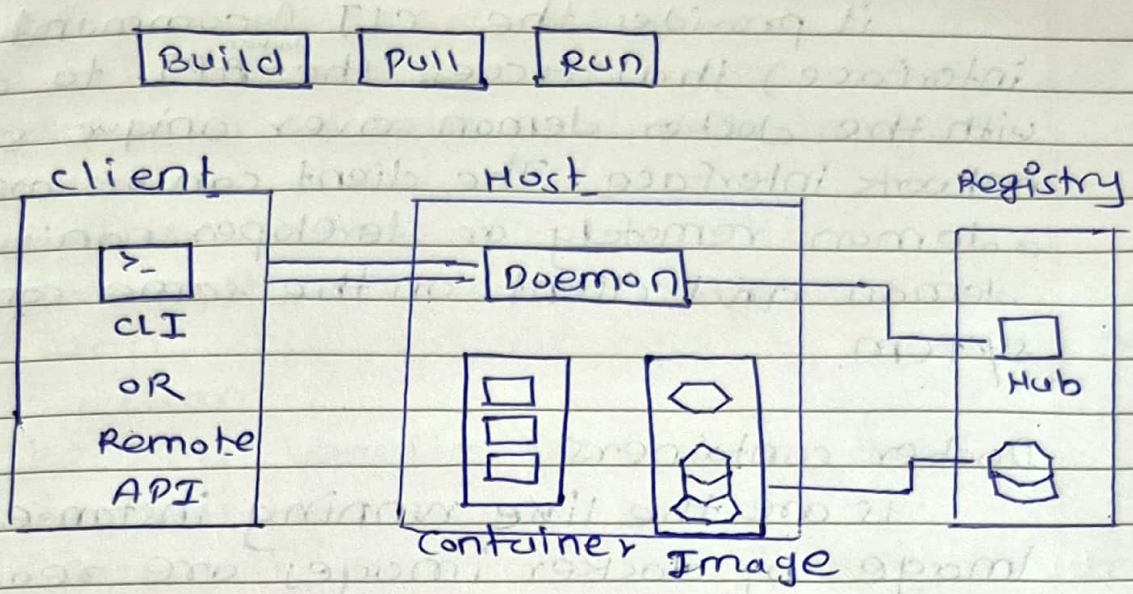
iii) Slower startup
Take longer to boot due to the need to load the guest os.

- Doccur

What is doccur?
A doccur is a open source platform that enables developers to package application and there dependancies into portable containe

that can run consistantly across various environment.

Docurr Architecture.



Docker usage client server architecture following is a break down of the core component associated with docker.

Docker Host

A Docker Host is a physical or virtual machine Running linux or another docker engine compatable os.

Docker Engine

Docker engine is a client server application consisting of the docker doemon. a docker API that interacts with the doemon and command line interface that docs to the doemon.

Docker Doemon

is a service that creates and manages docker images. by using the commands from the client

essentially the docker daemon search as a control centre for the docker implementation.

Docker client
it provider the CLI (command line interface) that access the docker API to communicate with the docker demon over unique socket or network interface. The client can be connected to a demon remotely or developer can run the demon and client on the some computer system.

Docker containers
is are the like running instances of docker image by docker images are read only files

Docker image
A Docker image contain executable appln source code and all the tools, libraries and dependencies the appl code needs to run as a container.

Docker build
is a command that has tools and features for building docker image.

Docker files
each docker container starts with the simple text file containing instructions for how to build docker container image.

Docker Hub
is the public repository of docker images

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docker hub includes images produce by docker certified images belonging to the docker Trusted registry and thousands of other images.

Docker Registry
is a scalable, open source storage and distribution system or docker image.

- Benefits of Docker
- Docker has revolutionerize the way we develop, ship and run appli^n by enabling containerization
- The key benefits are docker are:

1) Portability
- consistant environment: docker ensures that application run the same way across different environments.
- cross platform compatibility: docker containe can run on any system that supports docker.

2) Efficiency:
- lightweight: docker containers shares the host system os kernal making them more light weight.
- fast startup time: container can be started and stop quickly.

3) Scalability & performance:
- efficient resource utilization: unlike vm's containers do not require a full os to be replecated.

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• easy scaling: scalling can be done easily by running multiple instance of container.

4) Isolation
• Application isolation: each application runs in own isolated environment and do not interfare with each other.
• process Isolation: processes running inside a container are isolated from those running own the host or other containers.

5) Continues integration & continues deployment
• Integration with CI/CD tools
• Rapid and reliable deployment

• Microservice architecture
service decupling: docker facilities development and deployment of microservices by allowing each service to run in its own container.

6) Security
containers provide a level of security through isolation ensuring that application and their dependancy are reparated from the host system and each other.

• KUberNetice
it is the backbone of model contain orchastration enabling scalable, resilant and automated application deployment.
it is an open source orchastration tool developed by google.

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It helps, manage containerize applications in different deployment environment physical virtual and cloud.

- Features
  - High availability in down time
  - scalability or high performance
  - disaster, recovery backup and restore

- Components
  1) Pods : smallest unit of ubernatice
  - Abstraction over container
  - each pods has its own ip address
  - There is one appl per pod

2) Service
  - each appl is in service which has a permanent ip address
  - lif cycle of pod and service are not connected
  - There are two type of services
  1) external for an application so we can access it through the browser
  2) internal for database that shouldn't be exposed

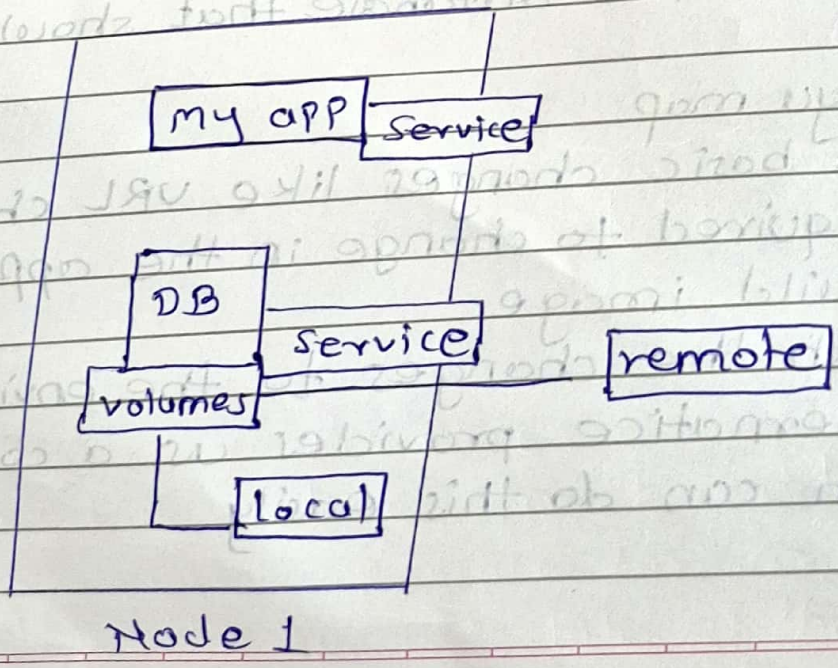
3) Configu map
  - For basic changes like URL changes could be required to change in the appl properties
  - Rebuild image
  - Push the changes to the environment
  - ubernatice provides us a configu map which can do this easily

4) secret : it is use to source secret data it is base 64 encoded

5) volumes: if your database crashes data is lost. ubernatice doesn't manage data persistancy. we can use volumes as local machine or external hard device drives. we as users are responcible for persistancy and backup

6) deployements: if one database or appi crashes their could be down time. in this case we need to specify the number of backup or container we want.
- The service we will point the same ip address.
- Helps in load balancing.

7) Stateful set
we can deploye data base directly because multiple containers could be writing to the same database for this we manage a stateful set.



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- Kubernetice Architecture

Worker processes:
we need three process for the worker machine in ~~Kubernetice~~ to work

- kubelet

The kubelet is the primary node agent that ensures containers are running in a pod as specified by the kubernetice control plane.

1) Functions:
- it receives instructures from the API server an ensures that the container describe in those specifications are running and healthy on its node.

2) interaction:
in the kubelet interact with the container runtime the container runtime interface manage container lifecycle image pooling and health checks.

3) Role: it reports the states of the node and its running pods back to the control plane allowing the cluster manager to make inform decisions about resource allocation and sheduling

- kubeproxy

is a network proxy that runs on each node to implement the kubernetice service concept.

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1) Function: it maintance network rules to allow network communication to and from Pods within the cluster.

2) Role: it manages port forwarding and simple load balancing for services, ensuring that traffic to a stable service ip is routed to the correct dynamically changing routed pod ip

- Container runtime

The container runtime is the underline software that is responsible for executing containers and managing containe image on node.

function:
it pulls container image from a registry unpack them and runs the appln within an isolated environment managing resources like CPU and memory. using mechanisms like C groups and namespaces

Role:
it provides a necessary environment for the application workload to run efficiently and securly

- MasterNode & Processes:

The kubernetice masternode runs several four processes responsible for managing and maintaining the desired state of the entire cluster

This components at as the brain of the cluster while worker nodes runs the actual applications. The main masterprocesses are.

i) API server.

This is the frontend of the kubernetice control plane and the single entry point for all communication it exposes the kubernetice API, process reset request, barrigets them and stores the configuration data in etcd

ii) etcd

A consistant highely available key value store that acts as the clusters single source of truth. it stores all data configuration and the current state of the cluster such as node detail services and node status.

iii) shedular

The process forcher for newly created node that have no node assign and selects an appropriate node for them to run over on it makes decision based on various factors like resource requirements, hardware constain, policy and availability.

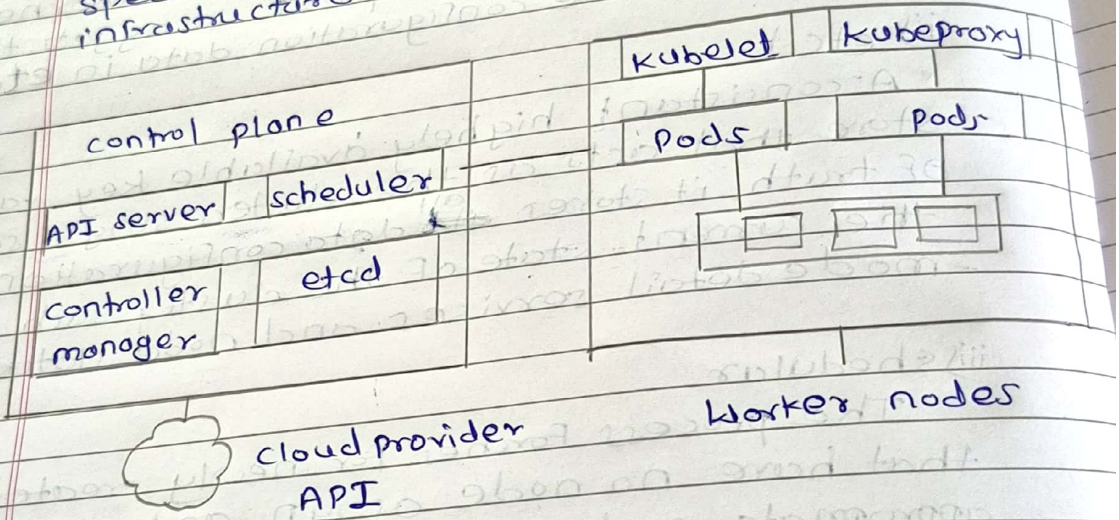
iv) controlel manager

kube controler manager is demon that runs various control loops that continuesly monitor the actual state of the cluster the api server and work to more it towards the desired state.

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cloud controller manager:
This component is used in cloud environments to manage interaction with the underline cloud providers API for task such as managing load balancers and storage volumes. it only runs controls specific to the cloud provider allowing cloud infrastructure to integrit with kubernetes.



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